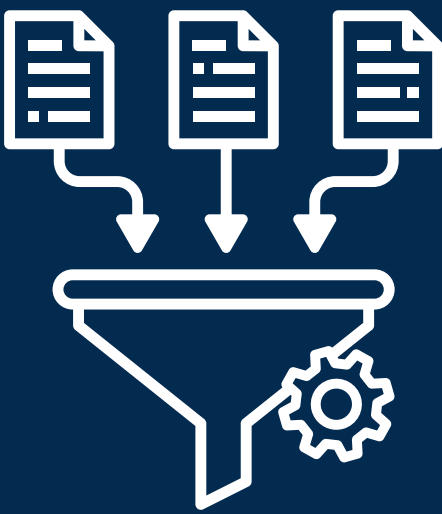




Discovering Hidden Patterns with ASSOCIATION RULE Mining



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INTRODUCTION

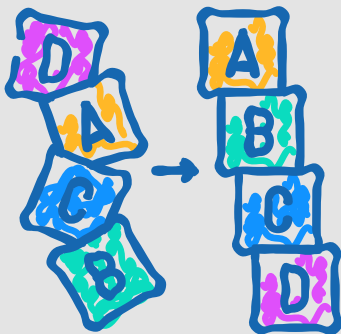


Association rules are if-then statements that uncover relationships between items in a dataset. They are commonly used in data mining to identify patterns, such as products frequently bought together.

Purpose & Context:

- Help businesses understand customer behavior.
- Improve decision-making in marketing, inventory, and recommendations.

KEY METRICS: SUPPORT, CONFIDENCE & LIFT



Support:

The frequency of transactions that contain a particular itemset.

$$\text{Support}(A \rightarrow B) = \frac{\text{Transactions with both A and B}}{\text{Total transactions}}$$

Confidence:

The likelihood that item B is purchased when item A is purchased.

$$\text{Confidence}(A \rightarrow B) = \frac{\text{Support}(A \cup B)}{\text{Support}(A)}$$

Lift:

Measures how much more likely B is bought when A is bought compared to when B is bought alone.

$$\text{Lift}(A \rightarrow B) = \frac{\text{Confidence}(A \rightarrow B)}{\text{Support}(B)}$$

REAL-WORLD APPLICATION

Market Basket Analysis

In retail, association rules help find that customers who buy bread and butter often also buy jam. This helps stores place related items nearby to boost sales.



Healthcare Diagnosis

In hospital, influenza can be detected through observation of the the symptoms such as fever, cough, and fatigue. This patterns finding helps in early disease detection and treatment planning.



EXAMPLE DATASET AND MULTIPLE RULES



Transaction Table

TID	Items
T1	Milk, Bread
T2	Milk, Diaper, Beer, Eggs
T3	Milk, Diaper, Beer, Coke
T4	Bread, Milk, Diaper, Beer
T5	Bread, Milk, Diaper, Coke

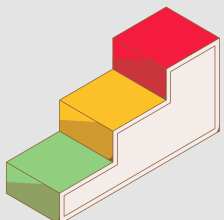
Frequent Itemsets (Support ≥ 0.6)

- {Milk} – 5/5 = 1.0
- {Diaper} – 4/5 = 0.8
- {Beer} – 3/5 = 0.6
- {Bread, Milk} – 3/5 = 0.6
- {Diaper, Milk} – 4/5 = 0.8
- {Diaper, Beer} – 3/5 = 0.6

Sample Rules (with Metrics)

Rule	Support	Confidence	Lift
Diaper → Milk	0.8	1	1
Diaper → Beer	0.6	0.75	1.25
Bread → Milk	0.6	1	1

SUMMARY STEPS OF APRIORI ALGORITHM



- 1.Set minimum support and confidence thresholds.
- 2.Generate candidate itemsets
- 3.Prune infrequent itemsets
- 4.Form frequent itemsets
- 5.Repeat with larger itemsets
- 6.Generate association rules
- 7.Evaluate rules using confidence and lift.